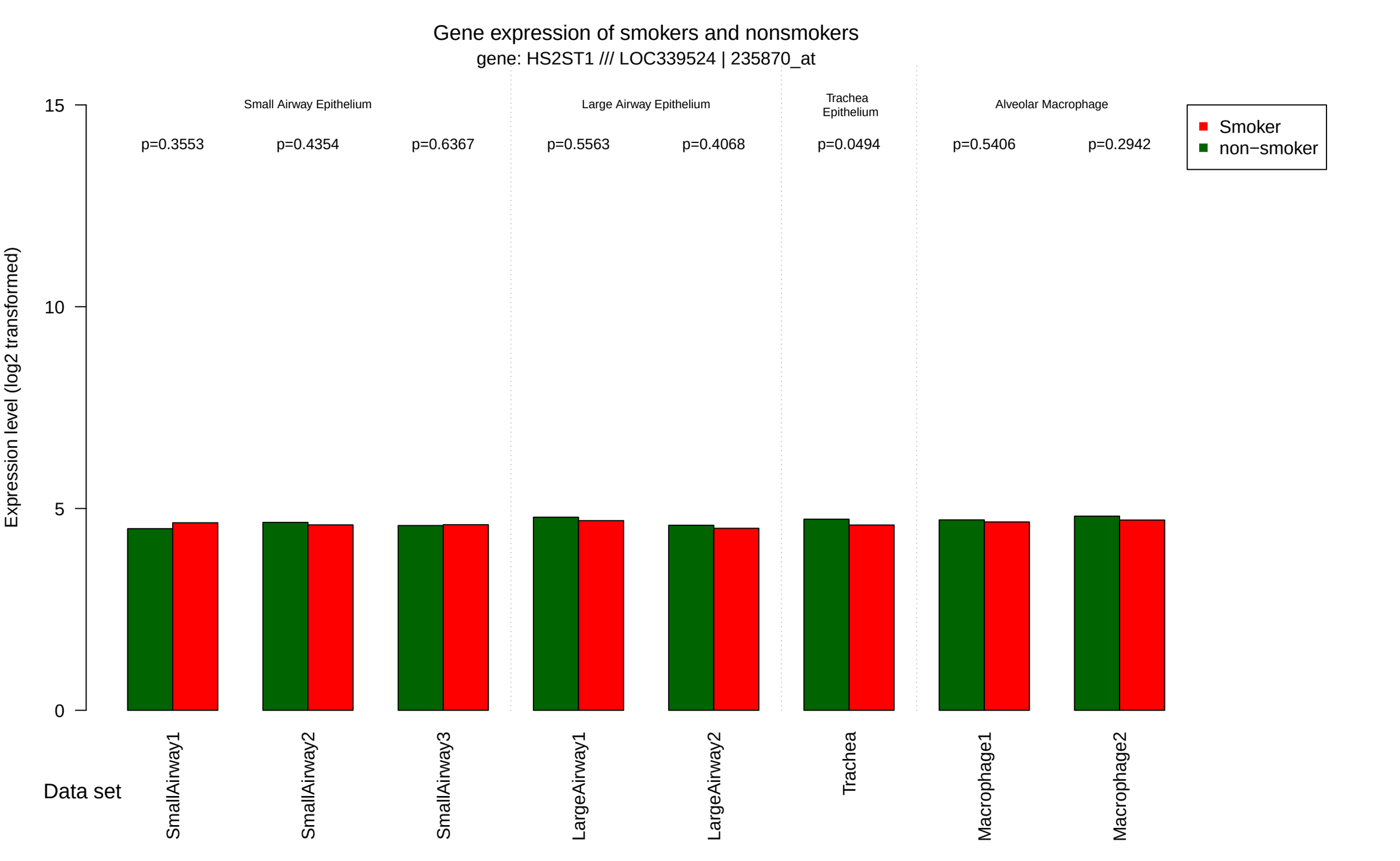
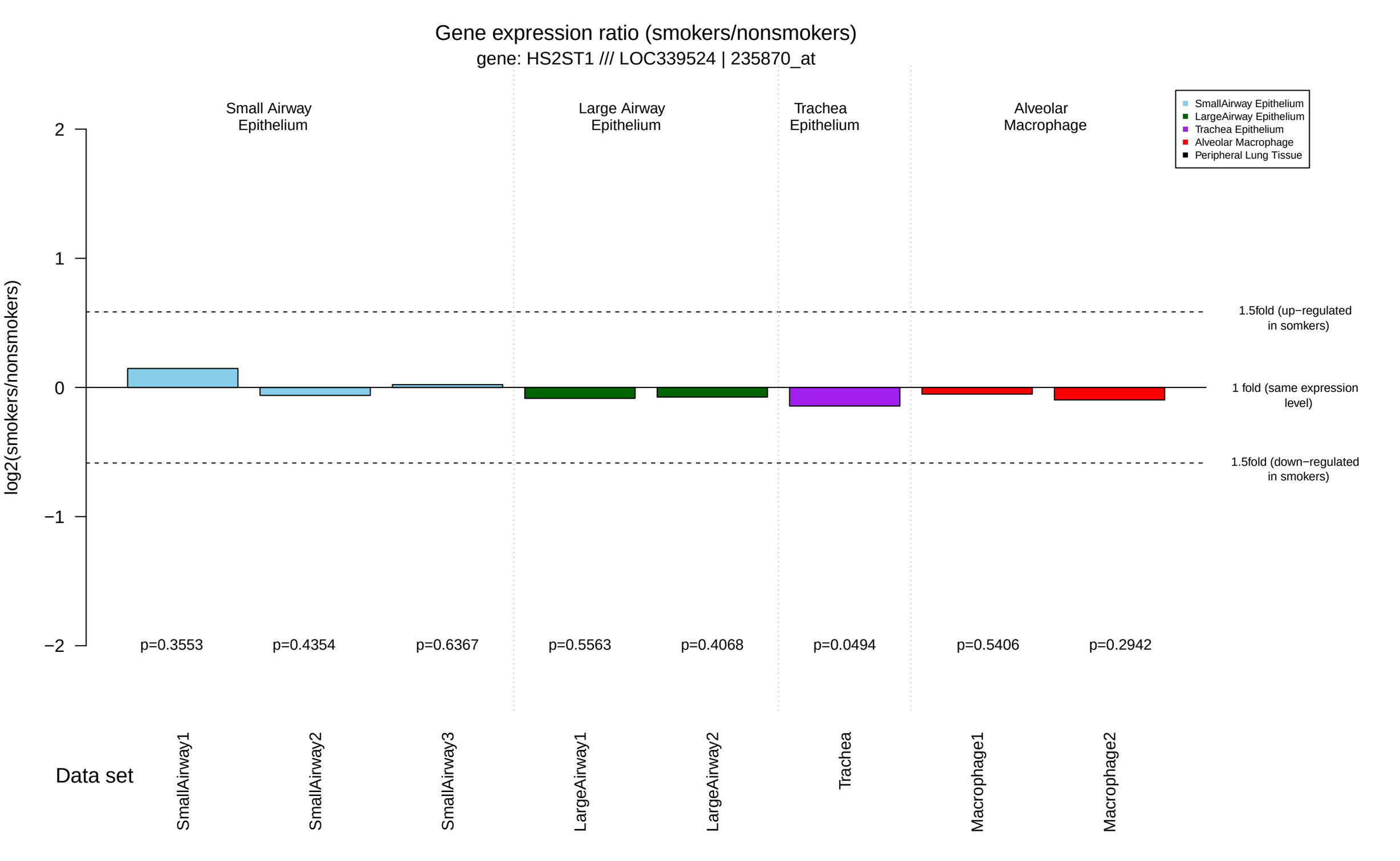
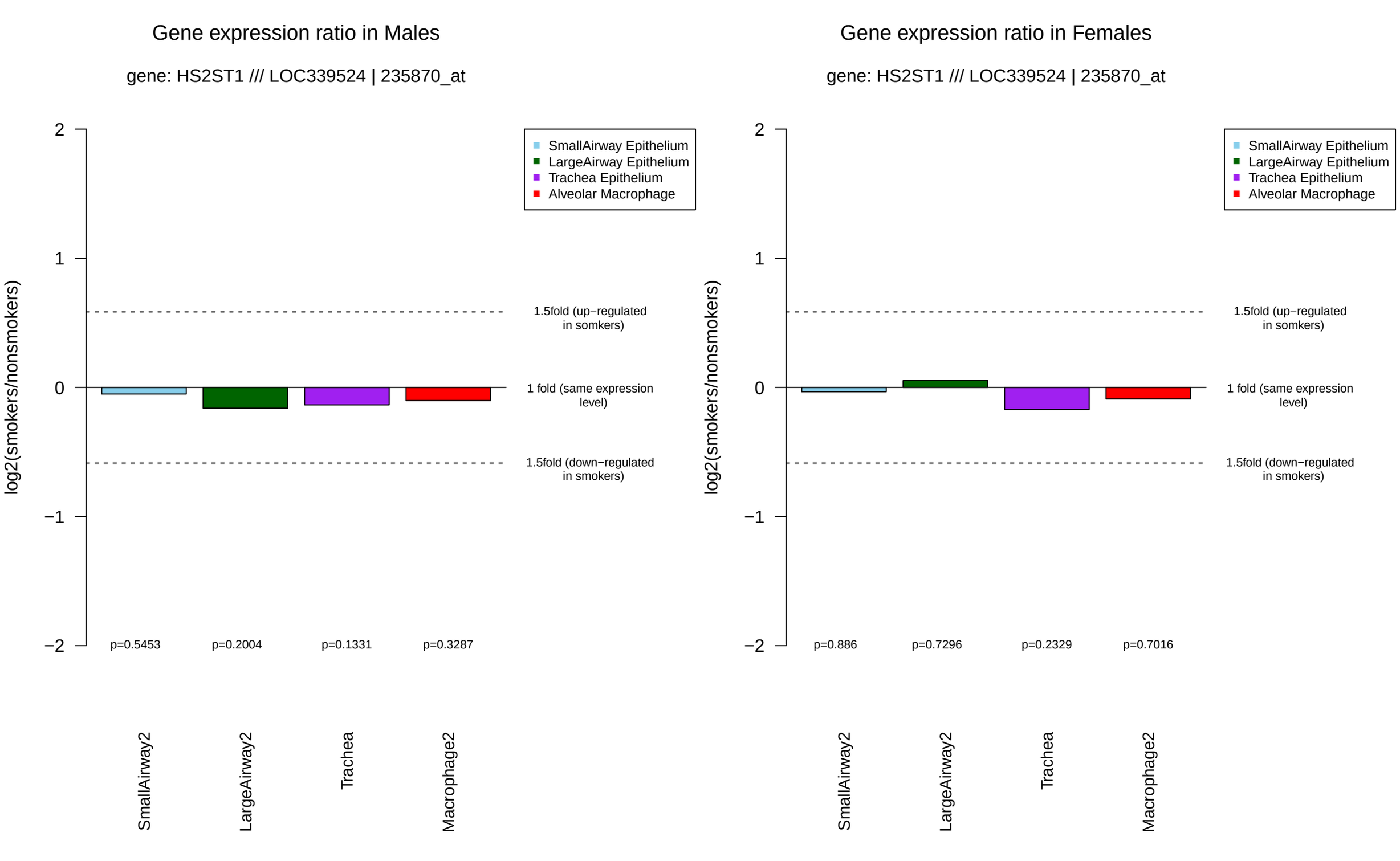
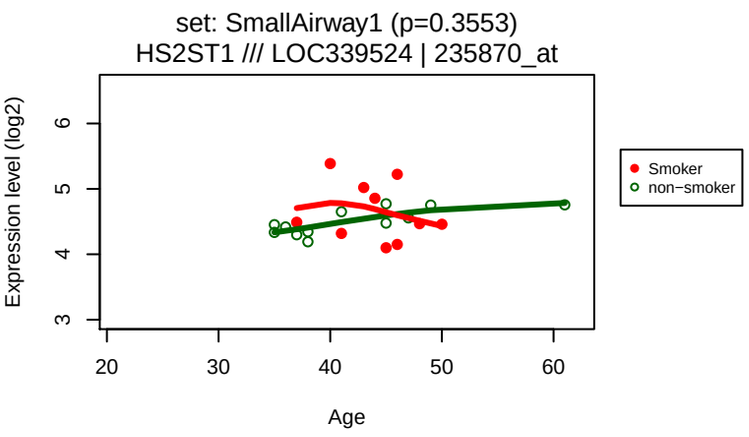
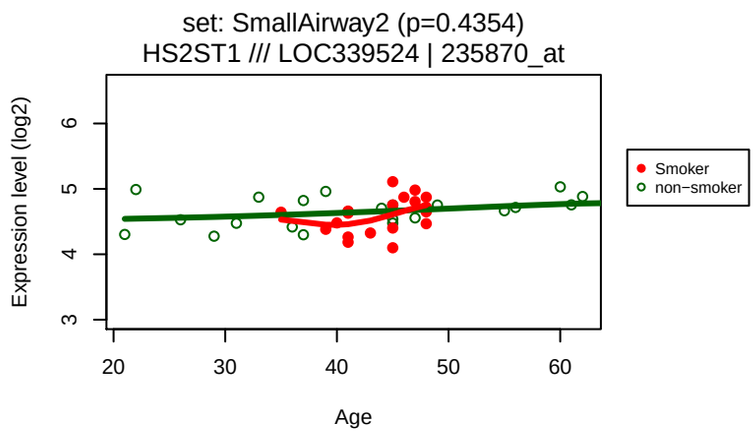
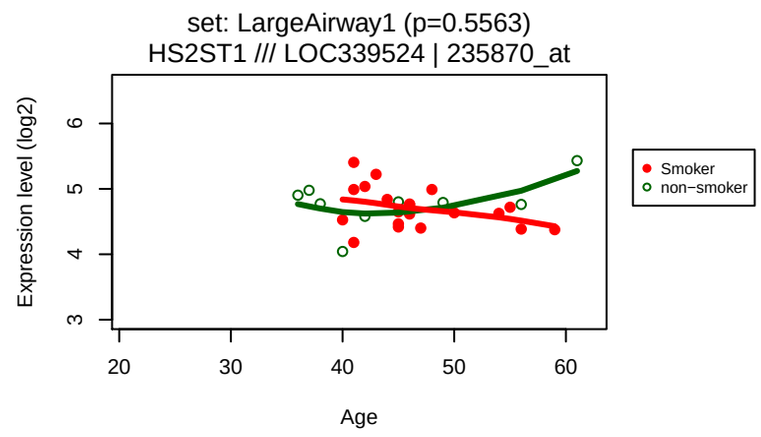

Gene expression of smokers and nonsmokers
gene: HS2ST1 /// LOC339524 | 235870_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3553
p=0.4354
p=0.6367
p=0.5563
p=0.4068
p=0.0494
p=0.5406
p=0.2942
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

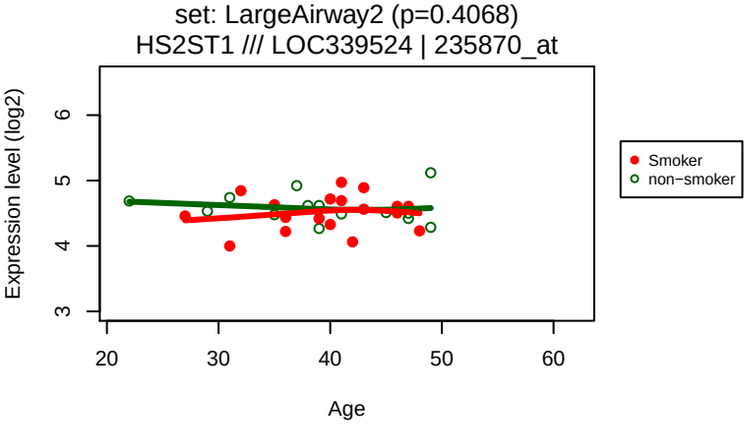

Gene expression ratio (smokers/nonsmokers)
gene: HS2ST1 /// LOC339524 | 235870_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3553
p=0.4354
p=0.6367
p=0.5563
p=0.4068
p=0.0494
p=0.5406
p=0.2942
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

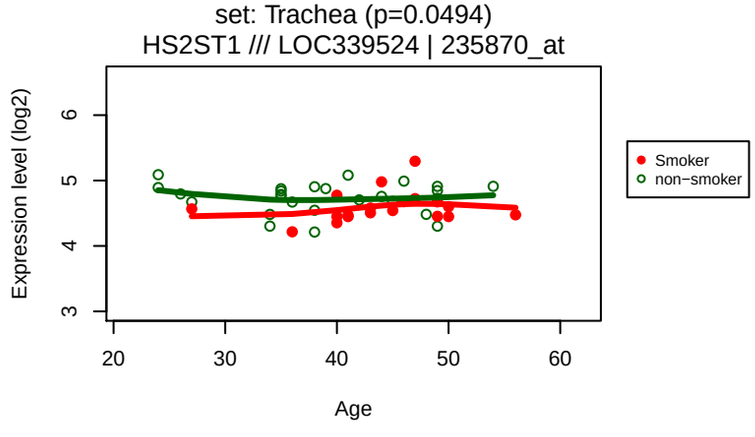

Gene expression ratio in Males
gene: HS2ST1 /// LOC339524 | 235870_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5453
p=0.2004
p=0.1331
p=0.3287
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HS2ST1 /// LOC339524 | 235870_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.886
p=0.7296
p=0.2329
p=0.7016
SmallAirway2
LargeAirway2
Trachea
Macrophage2

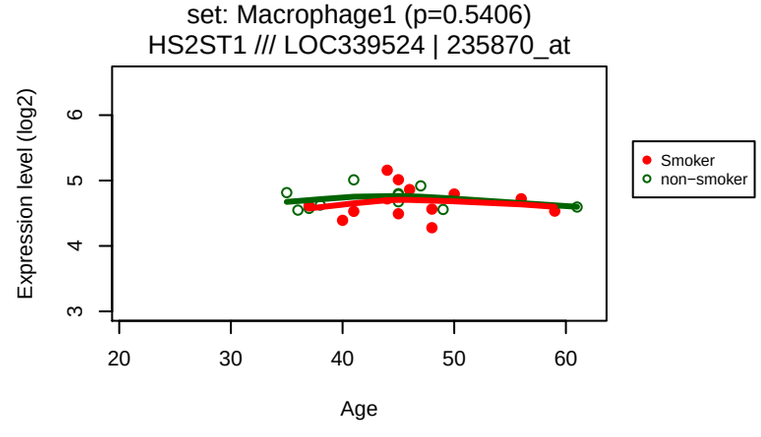

set: SmallAirway1 (p=0.3553)
HS2ST1 /// LOC339524 | 235870_at
Expression level (log2)
Age
Smoker
non−smoker

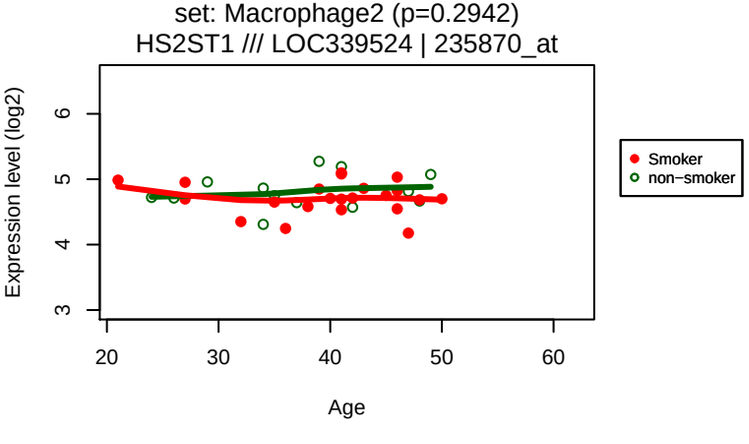

set: SmallAirway2 (p=0.4354)
HS2ST1 /// LOC339524 | 235870_at
Expression level (log2)
Age
Smoker
non−smoker

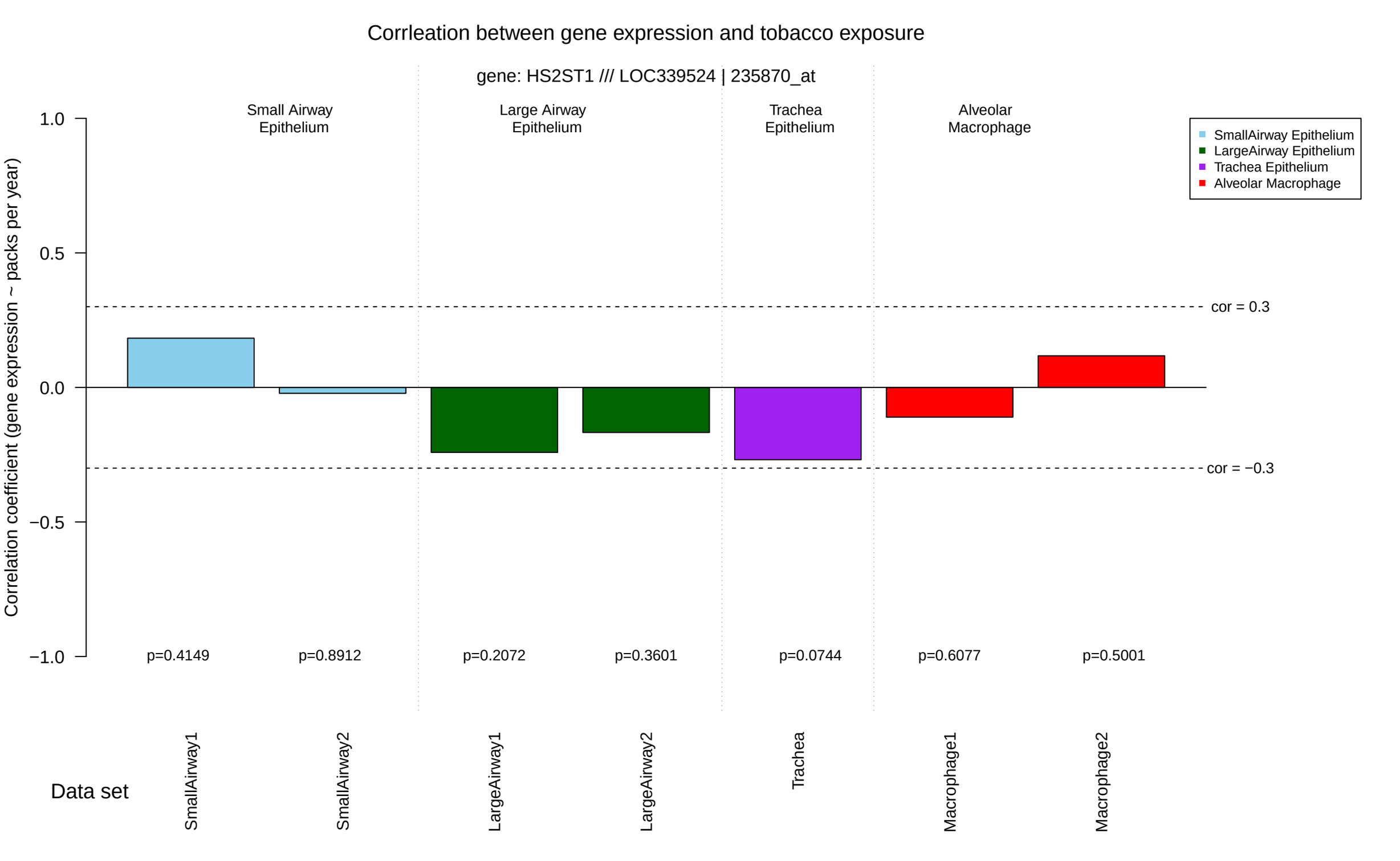

set: LargeAirway1 (p=0.5563)
HS2ST1 /// LOC339524 | 235870_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4068)
HS2ST1 /// LOC339524 | 235870_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0494)
HS2ST1 /// LOC339524 | 235870_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5406)
HS2ST1 /// LOC339524 | 235870_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2942)
HS2ST1 /// LOC339524 | 235870_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HS2ST1 /// LOC339524 | 235870_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4149
p=0.8912
p=0.2072
p=0.3601
p=0.0744
p=0.6077
p=0.5001
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2